



Detailed and simplified models for evaluation of effective thermal conductivity of open-cell porous foams at high temperatures in presence of thermal radiation



Miguel A.A. Mendes^{a,*}, Prabal Talukdar^{b,*}, Subhashis Ray^{a,*}, Dimosthenis Trimis^a

^a Institute of Thermal Engineering, Technische Universität Bergakademie Freiberg, Gustav-Zeuner-Strasse 7, D-09596 Freiberg, Germany

^b Department of Mechanical Engineering, Indian Institute of Technology Delhi, New Delhi 110016, India

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ABSTRACT

Simulations of coupled conduction–radiation heat transfer through open-cell porous foams have been performed in this article in order to evaluate the effective thermal conductivity (ETC) at high temperatures due to combined effects. For this purpose, the complex structure of foams has been generated using the knowledge obtained from 3D Computer Tomography (CT)-scan images and both detailed and simplified homogeneous modes have been employed during the present investigation. For detailed models, in addition to the energy conservation equation, the radiative transfer equation has been solved in order to obtain the distribution of radiation intensity using the blocked-off region approach, which can handle the voxel information acquired from the CT-scan data. Proposed simplified models, on the other hand, are based on homogenization approach and the performance of these simplified models has been compared with the reference model in terms of their accuracy. Simplified models mainly require information about the ETC due to pure conduction calculated from detailed 3D simulation and the extinction coefficient estimated from the CT-scan data using the image processing technique. Extensive parametric study has been conducted and the analysis of results show that ETCs estimated from the simplified models reasonably agree with those obtained from the most accurate model, depending upon the chosen value of extinction coefficient.

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1. Introduction

Developmental efforts on producing varieties of ceramic foams have increased significantly in the recent years due to their ever-increasing applications in numerous technological fields. The unique structural properties, such as large surface area to volume ratio of metal/ceramic foams (porous media), makes it very useful for many industrial applications, especially where heat transfer is of primary concern. In this respect, the evaluation of effective thermal conductivity (henceforth will be referred to as the ETC, throughout the paper) of porous foams has become a major requirement for accurate prediction of heat transfer behaviour of many systems, such as heat exchangers, thermal insulators or combustion systems, to name a few. The complex morphology of foam structures plays an important role in the heat transfer characteristics and consequently, the ETC can be quite different for different

foams, even with same porosity. If the heat transfer takes place at a high temperature, the ETC has a larger contribution from thermal radiation and hence an accurate modelling of radiation heat transfer becomes inevitable.

Unfortunately, however, the morphology of porous media is generally quite complex and resolving these structures with a numerical model has always been a challenging task. Available literature on modelling of heat transfer through porous foams can be divided broadly into two categories, (i) a detailed model, considering the complex structure of the foam up to a certain resolution and (ii) use of a homogeneous model with effective thermo-physical and optical properties, calculated using some form of correlations or directly measured from experiments. Nowadays, a wide variety of approaches can be found under these two main classifications, which are discussed in the following paragraphs.

As far as detailed models are concerned, since actual foam structures are quite complex in nature, several researchers attempted to synthetically generate some equivalent regular structures that can reasonably approximate real foams for the purpose of heat transfer simulations. As reported in the literature, ceramic and metallic foam structures can be represented, for example, with cubic [1,2], dodecahedral or tetrakaidecahedral structures. It was

* Corresponding authors. Tel.: +49 3731 393946 (M.A.A. Mendes), +91 11 26596337 (P. Talukdar), +49 3731 393947 (S. Ray).

E-mail addresses: miguel.mendes@iwtt.tu-freiberg.de (M.A.A. Mendes), prabal@mech.iitd.ac.in, prabal.talukdar@gmail.com (P. Talukdar), ray@iwtt.tu-freiberg.de, juhp_sray@yahoo.co.in (S. Ray).

Nomenclature

G	incident radiation, W/m^2
I	radiation intensity, $\text{W/m}^2 \text{ sr}$
k	thermal conductivity, W/m K
L	length, m
$q, \vec{q}$	heat flux, heat flux vector, W/m^2
$\vec{r}$	position vector, m
s	distance along the direction of radiation intensity, m
$\hat{s}$	direction vector for radiation intensity
S	source function, W/m^3
T	temperature, K
x, y, z	cartesian coordinates, m

Greek symbols

β	extinction coefficient, $1/\text{m}$
ΔT	temperature difference between hot and cold boundaries
ε	emissivity
ϕ	azimuthal angle, rad
Φ	scattering phase function
κ	absorption coefficient, $1/\text{m}$
θ	polar angle, rad
Θ	scattering angle, rad
ρ	reflectivity
σ	Stefan–Boltzmann constant = $5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$

σ_s	scattering coefficient, $1/\text{m}$
τ	transmissivity
ω	scattering albedo
Ω	solid angle, sr

Subscripts

av	average
b	blackbody
c	cold
C	conductive
eff	effective
f	fluid
h	hot
max	maximum
min	minimum
r	representative
R	radiative
s	solid

Abbreviations

ETC	effective thermal conductivity
FVM	finite volume method

observed that dodecahedral and tetrakaidecahedral structures [1] could reasonably represent ceramic or metallic foams if some of the global properties, like porosity, pore density (pores per inch, in short, ppi), etc., are maintained. Kuhn et al. [3] used infinitely long and randomly oriented cylinders in order to describe struts of polyurethane foams. The unit cell was assumed as a pentagonal dodecahedron and triangular strut cross-sections were converted into their circular counterparts with the same geometric mean cross-sectional area. However, it was later realised on from the Computer Tomography (CT)-scan images that cells are not only made of pentagonal dodecahedron shapes but also tetrakaidecahedral cells do exist. Subsequently, it was confirmed by Dillard et al. [4] who showed that the most representative cells in polyurethane foams contain 12 faces. Most of the faces of their foams were pentagonal (57%) with the presence of considerable fractions of quadrilateral (17.6%) as well as hexagonal (21.6%) faces. Nevertheless, in terms of modelling effort, the construction of grid for these idealised structures, although not simple, is relatively easier than that for the actual foam structures.

In recent days, however, owing mainly to the availability of three-dimensional (3D) CT-scan imaging, researchers have started developing models using the real structural data of foams in digitised form. Although these geometric models are more accurate and hence realistic than modelled representative regular structures, they suffer from the requirement of significantly larger computational efforts. For example, regarding radiation heat transfer, Monte Carlo simulations [5] with these 3D CT-scan data could be computationally quite expensive for simulating these foams in practical applications. On the other hand, in order to simulate conduction heat transfer, although detailed simulations can be performed for small foam samples [6–8], in general, its modelling for significantly larger practical applications is also prohibitive and, to some extent, unjustified owing to the requirement of excessive computational time.

Homogeneous models, those use effective properties of porous media, can prove to be good alternatives for saving computational

effort. However, the main limitation with these models is to obtain the required properties in an accurate manner; in particular, the ETC due to conduction–radiation and other optical properties, like, extinction coefficient, scattering albedo and scattering phase function. Experiments are the common practice for the evaluation of these properties. Substantial amount of investigations regarding radiative properties [9–14] and the ETC due to pure conduction [15,16] have been reported in the literature along with their measured values. Monte-Carlo ray-tracings simulation is an attractive alternative for numerical evaluation of radiative properties and has been used by several researchers using the 3D CT-scan data for generating the true geometry of foam [17–20]. For example, in the work of Parthasarathy et al. [20], a ray-tracing Monte-Carlo simulation was presented. In their study, the computational grid was reconstructed from 3D CT-scan images and Magnetic Resonance Image (MRI)-scans of different reticulated porous media. A ray-tracing algorithm was adopted in order to track rays inside the grid structure. Statistically large number of rays was traced for their path-lengths and incident angles those were subsequently used for the determination of probability-based equivalent extinction coefficient and scattering phase function for various porous media with different porosities and pore densities.

Loretz et al. [21] discussed numerical methods for the determination of extinction coefficient along with its experimental measurements and reported reasonably good agreement between modelled predictions and the experimental data. Based on certain combination of pentagonal dodecahedral and/or tetrakaidecahedral cells, they subsequently proposed a relation for the extinction coefficient as a function of porosity and average cell diameter. Another new method was also proposed in their study, where extinction coefficients were determined based on the slice-by-slice superposition of 3D CT-scan images. They finally concluded that this new method, being quite easy to implement and computationally far less time consuming, also provides reasonable results since the method uses actual structure of foams in the form of 3D digitised data.

Regarding pure heat conduction through porous foams, several investigations can be found in the literature, where the detailed geometry with staircase approximation, obtained from the 3D CT-scan information of porous structures, were used [6–8]. For example, Mendes et al. [8] proposed a very simple yet efficient numerical method for the determination of ETC, considering conduction heat transfer through open-cell foam-like structures, based on the predicted ETC of the same medium under vacuum condition. From their results, four different correlations were proposed, covering the entire range of meaningful thermal conductivity ratio of fluid to solid less than unity. Their study clearly demonstrated that the ETC of open-cell porous foams under vacuum condition contains the necessary quantitative structural information as far as the effective thermal conductivity in presence of any working fluid is concerned in order obtain accurate correlation for a particular foam structure.

Very recently, Talukdar et al. [2] proposed a new methodology for solving radiative energy transport coupled with heat conduction through foam structures. A blocked-off region approach [22–24], integrated with the Finite Volume Method (FVM), was implemented in an in-house Fortran code in order to solve combined energy transport through porous media. The detailed geometry of the medium was represented in terms of voxel information. Two idealised structures, in the form of a 3D cubic cell with lumps and struts, as well as a 2D honeycomb section with uniform struts and without any lump, were investigated in their study although no real porous structure was considered. The method, however, proved to be simple and accurate to resolve the structure of a porous medium. In order to investigate the usefulness of this method, in the present study, the same methodology has been applied for a few real ceramic foam structures, where the voxel information has been obtained from 3D CT-scan images in order to represent the geometry of these foams with staircase approximation.

It will be shortly apparent that two detailed models have been used in the present study for evaluating the ETC of open-cell porous foams based on the information obtained from 3D CT-scan images. The first detailed model, considered as the reference, uses the methodology proposed by Talukdar et al. [2] without any further assumption. The second detailed model separately solves conduction and radiation heat transfer within the same geometry and finally the total ETC is calculated by adding individual contributions of ETCs for pure conduction and pure radiation. In addition to these detailed models, three homogeneous models have been also presented in this paper in order to compare the accuracy of all these models with respect to the reference model that solves the combined conduction–radiation problem, using the detailed geometry. The referred homogeneous models mainly require information regarding the ETC due to pure heat conduction and the extinction coefficient for radiation. Once again, these quantities have been evaluated using 3D CT-scan images.

One of the objectives of the present investigation is to compare the performance of simplified homogeneous models with results of detailed numerical simulations which are considerably time consuming. It is very important to study systematically the deviation of results obtained with these simplified models from detailed simulations in order to identify their applicability. Extensive parametric studies, therefore, have been carried out in order to compare and quantify their deviations under various operating conditions. To the best of authors' knowledge, such a comparison, considering the combined effects of conduction and radiation, has never been reported in the literature for porous media. The other objective is to study the performance of blocked-off region approach, proposed by Talukdar et al. [2] for real, open-cell, porous foams and to compare them with the homogenisation approach. Since the extinction coefficient, used for homogeneous models, has been evaluated in a

different manner from 3D CT-scan images, the comparison of detailed simulations with homogeneous models also indirectly proves that the methodology, proposed earlier for the detailed conduction–radiation model [2], is a reasonably accurate approach.

The present article is organised as follows. Section 2 deals with the mathematical formulation and the solution methodology. In that section, governing equations, 3D modelling with detailed geometry and calculation of effective thermal conductivity are presented in addition to reporting two detailed and three simplified homogeneous models along with the method for determining the extinction coefficient from 3D CT-scan images. Results are presented and discussed in Section 3 and finally, the conclusions from the present investigation are drawn in Section 4.

2. Mathematical formulation and solution methodology

2.1. Governing equations and 3-D modelling with detailed structure

Heat transfer phenomena within porous media are strongly influenced by its internal morphology, which, in real life, is extremely complicated. The information of the pore structure inside a porous medium can be adequately acquired using the 3D CT-scan imaging technique. The detailed model, used in this article, is capable of solving coupled conduction–radiation problem in any complex structure, described by the staircase approximation. In the previous article, results obtained using the same method were compared with those from the commercial software Fluent 6.3 for various regular structures and the comparison was found to be reasonably accurate [2]. The numerical model was essentially formulated using the finite volume method on Cartesian coordinates. The details of the method were described by Talukdar et al. [2] and hence only a brief description is presented here for the sake of completeness.

In the adopted approach, the following assumptions have been made in order to formulate the detailed model:

- (i) The size of representative samples of porous foams is large enough to represent macroscopic properties of the structure, for example, the representative size of samples L_r is typically chosen to be at least 3 to 4 times larger than the characteristic cell size of the matrix [18,20].
- (ii) Thermo-physical properties of the solid in struts and lumps, as well as the fluid (i.e., participating media) in free spaces (i.e., pores) are constant over the range of temperature encountered in simulations.
- (iii) Convection within the porous matrix and the thermal expansion of material are both negligible.
- (iv) Pore surfaces, i.e., solid surfaces in direct contact with the fluid are gray as well as diffuse, which is generally applicable for most of the ceramic and some of the metal foams.

However, it will be shortly apparent that variable thermo-physical properties can also be accommodated in the present formulation without any difficulty.

The steady state energy conservation equation, considering conduction and radiation heat transfer without any heat generation, is given by [5,25]:

$$\nabla \cdot (k \nabla T) - \nabla \cdot \vec{q}_R = 0 \quad (1)$$

where k is the local thermal conductivity, which in principle, being a material (solid or fluid) property, could also be a function of temperature and hence space. Further, $\vec{q}_R$ is the radiative heat flux vector and its divergence $\nabla \cdot \vec{q}_R$ is generally considered as the source term while discretising and solving Eq. (1) using FVM which has also been adopted in the present study. Discretisation procedure

of Eq. (1), along with the source term, is quite simple and hence is not presented here for the sake of brevity [26].

In Eq. (1), the divergence of radiative heat flux $\nabla \cdot \vec{q}_R$ requires the solution of radiation intensity, which has been obtained by solving the Radiative Transfer Equation (RTE). The RTE for a gray, absorbing as well as emitting medium in the direction $\hat{s}$ can be written as:

$$\frac{dI(\vec{r}, \hat{s})}{ds} = -\kappa(\vec{r})I(\vec{r}, \hat{s}) + S(\vec{r}, \hat{s}) \quad (2)$$

where the source function for an absorbing-emitting medium can be expressed as:

$$S(\vec{r}, \hat{s}) = \kappa(\vec{r})I_b(\vec{r}) \quad (3)$$

In Eqs. (2) and (3), $\vec{r}$ and $\hat{s}$ are the position vector and the unit vector describing the direction of radiative intensity, respectively, κ is the absorption coefficient of the medium and $I_b = \sigma T^4/\pi$ is the intensity of black body radiation, with σ being the Stefan-Boltzmann constant. Discretisation of the RTE has been carried out according to the suggestion of Chai et al. [27] and hence is not repeated here. During this process, 10×8 uniform control angles in polar (θ) and azimuthal (ϕ) directions, respectively, have been chosen, such that $\Delta\theta = 18^\circ$ and $\Delta\phi = 45^\circ$. Such a uniform division has been found to be sufficient for obtaining control angle independent solutions. For detailed information, readers may refer to some of the previously published article on this subject [2,27,28]. Once the distribution of radiation intensity has been obtained by solving Eq. (2) along with the source term in Eq. (3), the incident radiation G and the divergence of radiative heat flux $\nabla \cdot \vec{q}_R$ in the energy conservation equation (1) can be calculated as:

$$G = \int_{4\pi} I(\vec{r}, \hat{s}) d\Omega \quad (4a)$$

$$\nabla \cdot \vec{q}_R = \kappa[4\pi I_b - G] = \kappa(4\sigma T^4 - G) \quad (4b)$$

In connection with the radiative heat transfer with blocked-off region approach, control volumes that represent pores of the foam or spaces occupied by the fluid medium have been considered as 'active' control volumes, where solutions are required. The rest of the control volumes have been considered as 'inactive' control volumes, where solutions are really not meaningful since radiative heat transfer does not occur through the solid medium. Detailed implementation of the blocked-off region approach was described by Talukdar et al. [2] and hence is not presented in this article.

It will be shortly apparent that during the present investigation, some of the simplified models have been tested for their accuracy with respect to the detailed model without any simplifying assumption. For some of these models, conduction and radiation heat transfer have been decoupled from each other and hence solved separately. In such cases, when conduction is solved alone, $\nabla \cdot \vec{q}_R$ in Eq. (1) is dropped and as a result, Eqs. (2)–(4) are never invoked. On the other hand, when radiation is separately treated, Eq. (1) turns out to be redundant and the energy conservation equation is obtained by setting the right hand side of Eq. (4b) to zero. Eqs. (2)–(4a) are, however, required to be satisfied in order to solve the modified form of Eq. (4b) and the solution methodology for RTE remains the same.

2.2. Calculation of effective thermal conductivity

In order to evaluate the ETC of a representative cubic sample (of size $L_r \times L_r \times L_r$) of the porous structure in a particular direction, say x , all other boundary surfaces, parallel to the desired direction, have been set as adiabatic boundaries with no net (combined conductive and radiative) heat flux [2]. Constant temperatures have been specified at two opposite boundaries along x -direction, with

one of the boundaries considered at a higher temperature T_h and the other boundary has been maintained at a lower temperature T_c . The ETC of the sample of porous medium in x -direction has been then predicted from the averaged Fourier's law of heat conduction as follows:

$$k_{\text{eff}} = \frac{q_{\text{av}}}{\Delta T/L_r} \quad (5)$$

where, according to the present choice, L_r is also the length of the computational domain in x -direction, i.e., the distance between hot and cold boundaries and $\Delta T = T_h - T_c$ is the corresponding applied temperature difference. The total averaged heat flux q_{av} in x -direction has been calculated from converged solutions of numerical simulations. It should be noted here that Eq. (5) applies irrespective of whether conduction and radiation are solved separately or simultaneously. In case of separate simulations, however, the ETC due to either pure conduction or pure radiation is retrieved from Eq. (5).

2.3. 3D detailed models

The reference model, termed as Model-1 in this article, considers the detailed 3D geometry as described in Section 2.1 and the ETC has been calculated according to Eq. (5) as explained in the foregoing section. In this approach, combined conduction and radiation heat transfer has been solved within the porous medium by satisfying Eqs. (1)–(4). Another detailed 3D model that has been employed during this investigation is named as Model-2, which considers separate (i.e., decoupled) simulations for conduction and radiation heat transfer through porous media. For the part dealing only with heat conduction, 3D detailed simulations have been performed for both solid and fluid phases by solving Eq. (1), without the radiative source term $\nabla \cdot \vec{q}_R$, in order to calculate the conductive ETC of the medium. On the other hand, for the radiation part, separate detailed 3D simulations have been performed by solving only for the RTE, represented by Eqs. (2)–(4). Since conduction heat transfer has not been considered for this second part and hence there is no available mechanism for removal of heat from the solid material, in order to eliminate the possibility of global imbalance of energy, the interior solid boundaries of the porous structure that is in direct contact with the fluid medium had to be treated as adiabatic. The ETC due to pure radiation has been then calculated from the averaged radiative heat flux in x -direction as described before. Finally, the total ETC of the foam structure has been evaluated by adding up individual contributions of ETCs for pure conduction and pure radiation as follows:

$$k_{\text{eff}} = k_{\text{eff,C}} + k_{\text{eff,R}} = \frac{q_{\text{av,C}}}{\Delta T/L_r} + \frac{q_{\text{av,R}}}{\Delta T/L_r} \quad (6)$$

where $q_{\text{av,C}}$ is the average heat flux due to heat conduction in absence of radiation and $q_{\text{av,R}}$ is the averaged pure radiation heat flux when conduction heat transfer is neglected. It may be emphasised here that for both these models, no further simplifying assumption has been invoked with respect to the geometry of porous foam and it has been generated directly from 3D CT-scan images of porous media with staircase approximation.

2.4. Simplified homogeneous models

Other than the detailed models, described in Section 2.3, three simplified models have been also considered in this article. All these models are essentially based on homogenisation approach. Important parameters, required for these approaches, are the ETC due to pure conduction and the extinction coefficient for radiation. The ETC for pure conduction has been calculated from the detailed 3D simulation considering the real structure of the foam based on

3D CT-scan data. On the other hand, calculation of radiation heat transfer with one-dimensional (1D) homogenisation approach requires information of extinction coefficient β , scattering coefficient σ_s and scattering phase function Φ . The extinction coefficients of different foam samples have been calculated from 3D CT-scan data using the image processing technique proposed by Loretz et al. [21], which is discussed in Section 2.5. The scattering coefficient σ_s and the scattering albedo ω can be calculated from the reflectivity ρ of foam surface [1], which can be assumed as diffuse, opaque and gray for all ceramic foam samples considered in this study. The relationship between σ_s and ρ (or ω) under this assumption can be given as [5]:

$$\frac{\sigma_s}{\beta} = \omega = \rho \quad (7)$$

Moreover, using the consequence of Kirchhoff's law for opaque surfaces ($\rho + \alpha = 1$), the reflectivity can be easily obtained if the emissivity ε of the foam surface is known.

The first simplified model, denoted as Model-3, is a 1D homogeneous conduction–radiation model. According to this model, both conduction and radiation have been solved simultaneously using the one-dimensional form of Eq. (1). The inputs to this model are the ETC from the simulation of pure conduction and the extinction as well as the scattering coefficients. Moreover, for the scattering phase function, the result for large diffuse gray spheres [5] has been used, which is a reasonable assumption for foams considered in this study [20]. In terms of scattering angle Θ this can be expressed as:

$$\Phi(\Theta) = \frac{8}{3\pi} (\sin \Theta - \Theta \cos \Theta) \quad (8)$$

From the converged solution of this one-dimensional problem, the total ETC has been calculated according to the relation given in Eq. (5).

The second simplified model, referred to as Model-4, is also another homogeneous model. However, similar to Model-2, it separately calculates the ETCs for conduction and radiation. The ETC due to pure conduction has been calculated exactly in the same manner as mentioned for Models-2 and 3. The ETC for pure radiation has been calculated by solving the RTE (along with associated equations) for a gray, absorbing, emitting and scattering medium in one-dimension, which appears to be similar to Eq. (2) and can be written as [5]:

$$\frac{dI(\vec{r}, \hat{s})}{ds} = -\beta(\vec{r})I(\vec{r}, \hat{s}) + S(\vec{r}, \hat{s}) \quad (9)$$

where the source function for one-dimensional problem is now different from Eq. (3), containing an extra term on the right hand side, and can be expressed as [5]:

$$S(\vec{r}, \hat{s}) = \kappa(\vec{r})I_b(\vec{r}) + \frac{\sigma_s}{4\pi} \int_{4\pi} I(\vec{r}, \hat{s}_i) \Phi(\hat{s}_i, \hat{s}) d\Omega_i \quad (10)$$

For this model, the ETCs for pure conduction and radiation have been calculated individually and subsequently they are added in order to calculate the total ETC according to Eq. (6).

The RTE for both Models-3 and 4 has been solved using the discrete transfer method (DTM) [23,29,30]. For one-dimensional calculations, discrete transfer method is known to be quite economical as well as accurate and hence has been considered for the present investigation. The solution procedure with DTM for radiation problem [23,29,30] and combined conduction–radiation problem [31] can be found in the literature and hence is not presented here for the sake of brevity. One may also refer to the articles by Talukdar and Mishra [32,33] for further clarification on accuracy and implementation issues related to the DTM for anisotropic scattering.

In a manner, similar to Model-4, another simplified 1D model (named as Model-5) used in this work, separately calculates the ETC for conduction and radiation. However, in this model, the ETC due to pure radiation has been evaluated using the extremely simplified Rosseland approximation [5]:

$$k_{\text{eff,R}} = \frac{16\sigma T_{\text{av}}^3}{3\beta} \quad (11)$$

While calculating $k_{\text{eff,R}}$, the temperature T_{av} in Eq. (11) has been assumed as the arithmetic mean temperature at the two boundaries, T_h and T_c . The total ETC has been then calculated from Eq. (6). It may be mentioned here that in principle, for a combined conduction–radiation problem with Rosseland approximation, the following modified energy conservation equation in one-dimension should have been solved:

$$\frac{d}{dx} \left[\left(k_{\text{eff,C}} + \frac{16\sigma T^3}{3\beta} \right) \frac{dT}{dx} \right] = 0 \quad (12)$$

where T is the local temperature and $k_{\text{eff,C}}$ is the ETC due to pure conduction, explained earlier and defined in Eq. (6). It is quite obvious that since the local radiative conductivity is a function of the dependent variable T , iterative solution procedure for Eq. (12) would be required. From the converged solution, however, the total ETC due to combined effects of conduction and radiation should have been calculated according to Eq. (5). This procedure, which appears to be more logical and appropriate, has been tried during the present investigation and results of the ETC have been found to differ marginally from those obtained with the adopted approach for all considered temperature ranges. Therefore, results in this article are reported by calculating radiative ETC according to Eq. (11), while evaluating the total (or combined) ETC from Eq. (6).

Table 1 resumes the main features of both detailed and simplified models considered in this article. It may be noted here that Models 1 and 2 are heterogeneous models that separately take into account both solid and fluid phases, while Models 3 to 5 are homogeneous models, treating the combination of solid and fluid phases

Table 1
Essential features of different models used for the evaluation of ETC.

Model	Heat transfer mode	Modeling approach	Evaluation of ETC	Governing equations	Required parameters
1	Conduction Radiation	3D heterogeneous	Coupled, Eq. (5)	Eqs. (1)–(4)	k_s, k_f $\varepsilon, \kappa = 0, \Phi = 1$
2	Conduction Radiation	3D heterogeneous	Separated, Eq. (6)	Eq. (1) with $\nabla \cdot \vec{q}_R = 0$ Eqs. (2)–(4) with $\nabla \cdot \vec{q}_R = 0$	k_s, k_f $\varepsilon = \kappa = 0, \Phi = 1$
3	Conduction Radiation	1D homogeneous	Coupled, Eq. (5)	1D version of Eq. (1) Eqs. (4), (9), (10)	$k_{\text{eff,C}}$ from 3D heterogeneous conduction ε, β Eq. (13), σ_s Eq. (7), Φ Eq. (8)
4	Conduction Radiation	3D heterogeneous 1D homogeneous	Separated, Eq. (6)	Eq. (1) with $\nabla \cdot \vec{q}_R = 0$ Eqs. (9), (10), (4) with $\nabla \cdot \vec{q}_R = 0$	k_s, k_f ε, β Eq. (13), σ_s Eq. (7), Φ Eq. (8)
5	Conduction Radiation	3D heterogeneous 1D homogeneous	Separated, Eq. (6)	Eq. (1) with $\nabla \cdot \vec{q}_R = 0$ Eq. (11)	k_s, k_f β Eq. (13)

as an equivalent medium. For heterogeneous models, the scattering has been neglected since, in the present study, the fluid phase has been assumed to be a non-participating medium for radiation purposes. For homogeneous models, however, the equivalent phase has to act a participating medium where scattering should be present and is a consequence of the irregular reflection of radiation at interior solid surfaces.

2.5. Determination of extinction coefficient β

The extinction coefficient β of a foam sample has been evaluated using a simple technique proposed by Loretz et al. [21], where no further simplifying assumption is made. By taking a representative structure of the foam sample from 3D CT-scan images, β in a particular direction (either x, y or z) has been calculated along a thickness L from the measurements of cross-sectional area covered by fluid pixels and the total area of the sample perpendicular to the direction of measurement. The ratio of these two quantities is defined as the transmissivity τ of the foam structure. Using Beer's law, the extinction coefficient β can then be calculated as:

$$\beta = -\frac{\log(\tau)}{L} \quad (13)$$

It may be noted that β given by Eq. (13) is a function of the local coordinate perpendicular to the direction of measurement since the thickness L directly depends on the local coordinate. It will be shortly apparent that the procedure for evaluating extinction coefficient of the medium from 3D CT-scan data is quite accurate and hence could be trusted for further simplifying assumptions.

3. Results and discussion

3.1. Sample geometry from 3D CT-scan images

In the present investigation, ETC due to combined conduction and radiation heat transfer for three samples of different open-cell porous foams, named as Foam-1, Foam-2 and Foam-3, have been considered. The reconstructed geometries of these foams, as obtained from 3D CT-scan images, are shown in Fig. 1. Foams 1 and 2, presented in Fig. 1(a) and (b), respectively, have been obtained from 10 ppi Al_2O_3 -C ceramic foams with porosities of 0.57 and 0.74, respectively, and characteristic pore sizes of 3 mm and 4 mm, respectively [34]. Foam-3, as shown in Fig. 1(c), has been obtained from a 30 ppi Al_2O_3 ceramic foam with a porosity of 0.79 and a characteristic pore size of 2.8 mm [34]. For all these

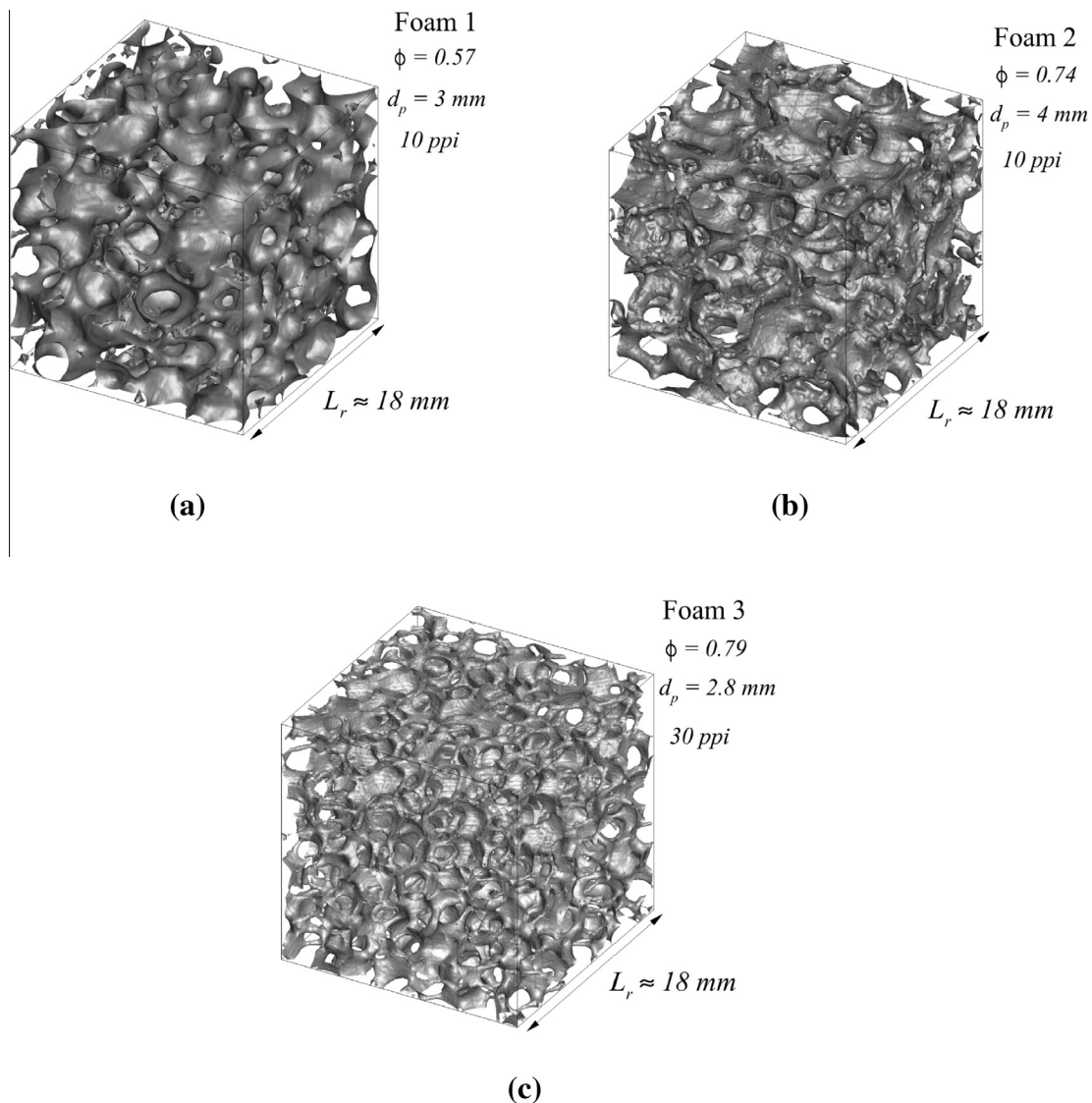


Fig. 1. Samples of different open-cell porous foams obtained from 3D CT-scan images: (a) Foam-1, (b) Foam-2 and (c) Foam-3.

three samples of ceramic foams, a representative sample size of $L_r \approx 18 \text{ mm}$ has been chosen and the corresponding CT-scan images have been obtained with a resolution of $65 \mu\text{m}$, except for Foam-1, where a resolution of $130 \mu\text{m}$ has been used. Therefore, for 3D detailed models described in Section 2.3, the grid size used in order to discretize the geometry of samples was around $140 \times 140 \times 140$ for Foam-1 and $280 \times 280 \times 280$ for Foam-2 and Foam-3. However, for simplified homogeneous models described in Section 2.4, grid sizes, considered for the 1D equivalent phase, depend only on the respective grid independence criteria.

3.2. Variations of extinction coefficient

Variations of β with respect to the non-dimensional thickness (made dimensionless with respect to the reference length L_r) in x , y and z -directions are shown in Fig. 2 for all three foam samples, described in the preceding section. It can be observed from the figure that, beyond a particular thickness, β has only a marginal variation in different directions for Foam-1 and Foam-2 and hence these two foams can be assumed nearly isotropic. However, for Foam-3, variations of β in different directions are observed to be significantly more pronounced. Since the ETC for detailed 3D models (i.e., Models 1 and 2) have been calculated along the z -direction, the value of β in the same direction has been considered for all the simplified homogeneous models (Models 3, 4 and 5) rather than assuming the averaged value of β for all three directions.

It may also be observed from Fig. 2 that β in a particular direction is, to some extent, non-uniform. The values of β have been found to be very large at smaller thicknesses that subsequently decrease very rapidly before exhibiting an arbitrarily oscillating

behaviour with low amplitude at higher values of thickness within the sample. This behaviour, however, is somewhat expected. It may be noted here that the porosity of a medium, i.e., the volume fraction occupied by the fluid phase, is quite similar to the fraction of cross-sectional area covered by fluid pixels in 2D CT-scan images. Therefore, for very small values of L (in the order of magnitude of the CT voxel resolution), the transmissivity τ will be nearly constant and similar in magnitude to the porosity. Consequently, according to Eq. (13), for extremely small thickness of the medium, the value of β will tend to infinity as L tends to zero. Nevertheless, since the initial values of β are substantially higher than that for the rest of the domain, while calculating β results for the initial part has been eliminated from the evaluation and an averaged value has been calculated based on the data for the remaining part of the sample. This observation, which is explained in before, is also in agreement with that reported by Loretz et al. [21] who pointed out that the measurement of β for an extremely small thickness of the medium can highly over predict true results. Apart from the averaged value, the maximum and the minimum values of β have been also considered in numerical simulations for each of the foams with homogeneous models in order to demonstrate the sensitivity of combined ETC on β . Table 2 shows the average, maximum and minimum values of β in the z -direction for all the three samples of foam.

3.3. Results for effective thermal conductivity

In this section, the performance of proposed detailed and simplified models has been examined for different parameters. Detailed models (i.e., Models 1 and 2) have been tested for real

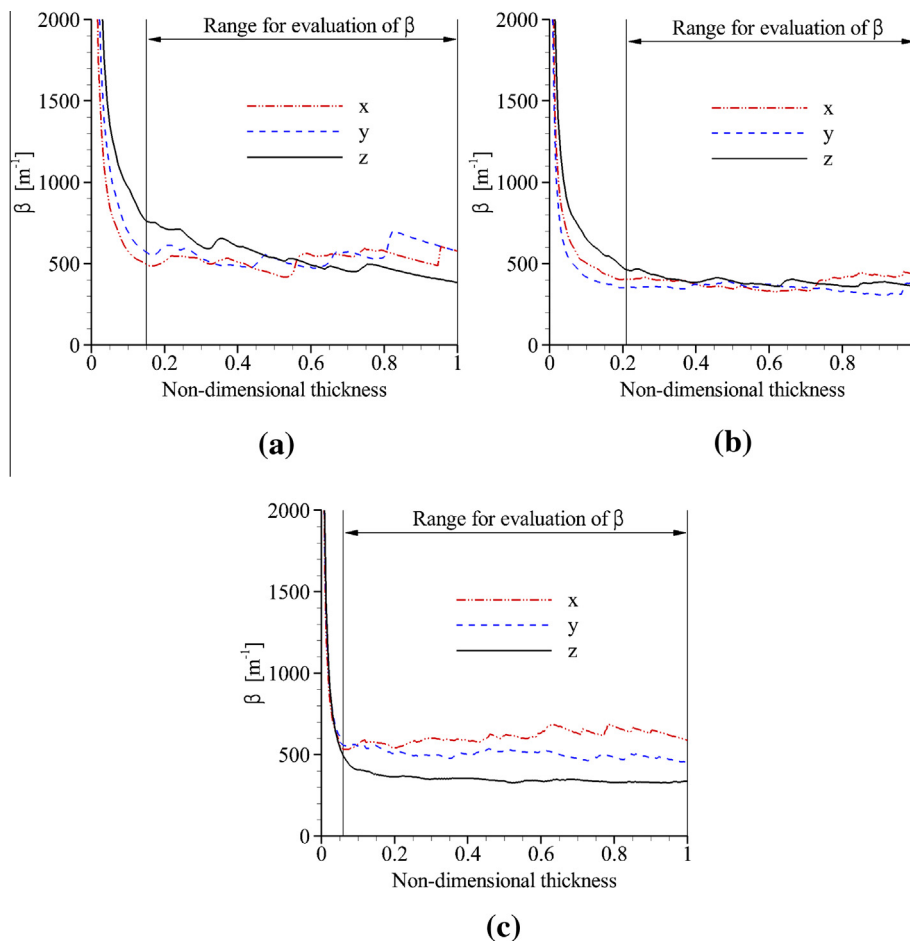


Fig. 2. Variations of extinction coefficient β as functions of dimensionless thickness x_i/L_r for (a) Foam-1, (b) Foam-2 and (c) Foam-3.

Table 2

Minimum, average and maximum values of extinction coefficient β in z-direction for three samples of porous foam.

Sample	$\beta_{\min}$ (m^{-1})	β_{av} (m^{-1})	$\beta_{\max}$ (m^{-1})
Foam-1	385	538	760
Foam-2	358	392	467
Foam-3	326	351	493

foams, whose resolved structures are presented in Section 3.1. As explained before, since these detailed models work with 3D geometries of foam structures and hence are computationally more expensive, the performance of alternative simplified, homogeneous 1D models has been also explored. These homogeneous models (i.e., Models 3, 4 and 5) have been also examined for various parameters and quantitatively compared with the reference model (i.e., Model-1), which can be considered the most accurate since it performs with the available detailed information about the porous foams and does not involve any further simplifying assumption.

Parameters considered for the present investigation are the average temperature of the domain T_{av} , taken as the arithmetic mean temperature of hot and cold boundaries, the applied temperature difference ΔT between these two boundaries, the emissivity ε of internal solid surfaces and the conductivity ratio k_s/k_f of the solid and the fluid medium. Table 3 presents all the cases chosen in this article for the parametric study. Case-1 has been considered as the reference case since parameters attributed for this case are based on typical values found in practice for high temperature applications with the use of ceramic foams. Based on this reference case, other relevant values of parameters are then selected such that they altogether represent a practical range of operating conditions involving high temperature applications.

The effective thermal conductivity for Foam-1 has been calculated using all five different models and results are shown in Figs. 3 and 4, where individual contributions of conduction and radiation are also indicated for models that calculate them separately (i.e., Models 2, 4 and 5). For homogeneous models (i.e., Models 3, 4 and 5), ETCs have been calculated using the maximum, the average and the minimum values of the extinction coefficient β presented in Table 2. Two extreme values of β , along with the averaged value, have been considered during the investigation in order to study the sensitivity of ETC with respect to β . This is all the more important to investigate this issue since the literature claims that determination of β from 3D CT-scan images tends to underestimate its measured value [21].

Fig. 3 shows the variation of ETC (in W/mK) with the average temperature using all the models, where subfigures (a)–(c) represent Cases 1–3, respectively, as listed in Table 3. As expected, the figure shows that with the increase in average temperature, the contribution of radiation in the ETC also increases. Since Fig. 3(b) represents the case with the largest average temperature (2000 K), the radiative contribution in ETC is the maximum for this case and it is the minimum for the case shown in Fig. 3(c), where

average temperature is the least (1000 K). The ETC, calculated by the reference model (Model-1) has been found to be 1.89, 3.22 and 1.06 W/mK for cases 1–3, respectively. With reference to Model-1, the ETC predicted by Model-2, underestimates its value approximately by 30% to 50% depending upon the contribution of radiation, which is a function of the average temperature. This clearly indicates that separate calculations of radiation, using 3D detailed model and subsequently adding results of the 3D detailed conduction model, may not produce the correct equivalent ETC as compared to the combined 3D conduction–radiation simulations. It is, however, worth mentioning that since in Model-2, radiation has been solved separately, adiabatic boundary condition had to be assumed at inner solid surfaces of the foam, which is possible only when these surfaces are purely reflecting ($\varepsilon = 0$). Therefore, for cases with large emissivity ($\varepsilon = 0.8$) of solid surfaces, as considered for cases 1–3, Model-2 fails to a considerably greater extent.

Variations of the ETC with respect to different parameters besides the average temperature, i.e., the temperature difference between hot and cold boundaries, the emissivity of solid surfaces and the conductivity ratio between the solid and the fluid medium, is presented in Fig. 4 for all the models. Parameters, considered for Cases 4–6, are listed in Table 3 and the corresponding results are shown in Fig. 4(a)–(c), respectively. Parameters for cases shown in Figs. 4(a) and 3(a) are the same except for the temperature difference ΔT , which has been taken as 200 K and 100 K, respectively for Cases 1 and 4. It can be clearly observed from these figures that the effect of ΔT on the ETC is negligible as long as the average temperature remains the same. Nevertheless, this outcome is based on the assumption that thermal conductivities of fluid and solid phases are independent of the operating temperature. Similarly, parameters for Cases 5 and 1, shown in Figs. 4(b) and 3(a), the only difference is the emissivity of internal solid surfaces of the foam, which has been taken as 0.2 and 0.8, respectively. It can be observed from these figures that a lower value of emissivity reduces the overall ETC of the foam and the performance of Model-2 clearly improves with respect to Model-1 with the decrease in emissivity. Fig. 4(c) represents a case with a large conductivity ratio of 1000 as compared to Case-1, shown in Fig. 3(a), where the conductivity ratio has been taken as 100. One can readily observe from Fig. 4(c) that the increase in conductivity ratio significantly increases the conduction heat transfer as compared to radiation. Therefore, accurate modelling of radiation does not exhibit a strong influence on the calculated ETC.

Based on the cases presented for Foam-1 in Figs. 3 and 4, it can be concluded that among three homogeneous models, Models 3 and 4 yield quite similar predictions for the ETC, whereas over-predictions are observed for Model-5 as long as the same value of β (either minimum, average, or maximum) is considered for all these models. Similar predictions obtained from Models 3 and 4 further justify the validity of superposition approach applied for homogeneous models for radiation with conduction. Such an observation was also reported earlier in the literature by Howell [25]. Nevertheless, a careful inspection of Figs. 3 and 4 further reveals that the ETC predicted by the combined simulation of conduction and radiation using Model-3 is always marginally higher than the result of superposition obtained with Model-4, probably owing to the complex interaction of conduction and radiation, which effectively enhances the heat transfer through the medium.

It is important to note that the ETC predicted by homogeneous models considerably depends on the chosen value of β . It can be observed from Figs. 3 and 4 that for all the considered cases use of $\beta_{\max}$ provides better agreement with the results obtained from Model-1, which has been selected as the reference model. This observation is somewhat expected since the present method of evaluation for β was reported to generally under-predict the value

Table 3

Different parameters considered in calculations of the ETC.

Cases	T_{av} (K)	ΔT (K)	ε	k_s/k_f
1 (Reference)	1500	100	0.8	100
2	2000	100	0.8	100
3	1000	100	0.8	100
4	1500	200	0.8	100
5	1500	100	0.2	100
6	1500	100	0.8	1000

Bold numerals represent change in parametric value with respect to reference case.

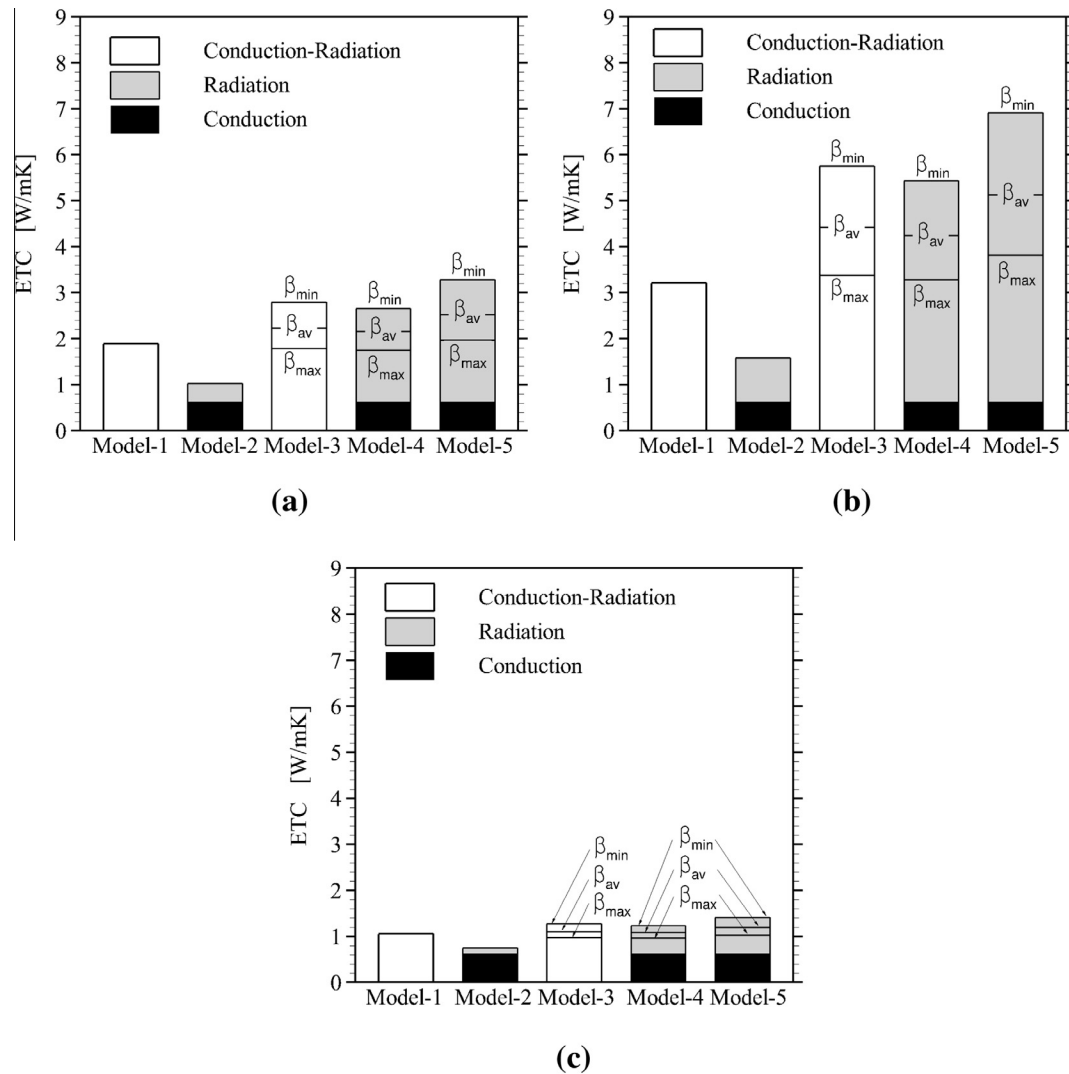


Fig. 3. Effect of average temperature T_{av} on the variation of ETC for Foam-1 obtained from different models: (a) Case-1 (reference) with $T_{av} = 1500$ K, (b) Case-2 with $T_{av} = 2000$ K and (c) Case-3 with $T_{av} = 1000$ K.

and hence β_{max} may be considered closer to the true value of β that would have been measured from the radiation experiment.

In an attempt to simplify the complex problem, many researchers apply the well-known Rosseland approximation for solving radiation problems and hence to estimate the ETC due to pure radiation. However, the Rosseland approximation is generally valid for optically thick media, which may not always be the case for open-cell porous foams. Hence, in order to check the validity and accuracy of the Rosseland approximation, in the present investigation, Model-5 has been introduced. It can be observed from Figs. 3 and 4 that generally speaking, the agreement between Models 5 and 1 is slightly inferior for all the considered cases in comparison to the agreement of Models 3 and 4 with Model-1. This is, however, also expected since Foam-1, with a porosity of 0.57, does not completely fall in the optically thick regime for all the operating temperature range. The optical thickness ($=\beta L_r$) of Foam-1 is found to be in the range of 7–14, depending upon the value of β presented in Table 2. The medium can be considered in the optically thick regime only if the optical thickness is more than 10 [5], which is marginally met only for higher values of β . Nevertheless, the Rosseland approximation yields very simple as well as explicit expression for the estimation of ETC due to pure radiation and hence can still be recommended if a fast solution is sought by sacrificing certain amount of accuracy.

Although, a qualitative comparison of different models can be realised from Figs. 3 and 4, more comprehensive and quantitative comparison can be inferred from the data in Table 4, which presents the relative error of ETCs predicted by Models 2 to 5 in comparison with the reference model for all the cases, listed in Table 3 for Foam-1. It can be observed from Table 4 that generally Models 3 and 4 provide the best predictions for the ETC if β_{max} is used in these models, yielding a relative error that is consistently lower than 10% for all the cases, except for Case-5. Results for this particular case have been obtained by considering an emissivity of 0.2 for the internal solid surfaces of the foam. Therefore, as may be expected, the contribution of scattering ($\omega = 1 - \epsilon = 0.8$) by the internal structure of the foam dominates the extinction of radiation and hence accurate modelling of the scattering phase function could play an important role. In the present study, for homogeneous models, a simplified scattering phase function, pertaining to large diffuse sphere, has been used which probably may lead to larger relative errors of approximately 15% for Case-5.

In order to test the performance of homogeneous models (Models 3 to 5) as compared to the detailed reference model (i.e., Model-1), two additional open-cell porous foams (Foams 2 and 3) have been also considered in the present investigation. Fig. 5(a) and (b) show the ETCs for Foams 2 and 3, respectively, with parameters considered for Case-1 (see Table 3 for further details). Model-2 has

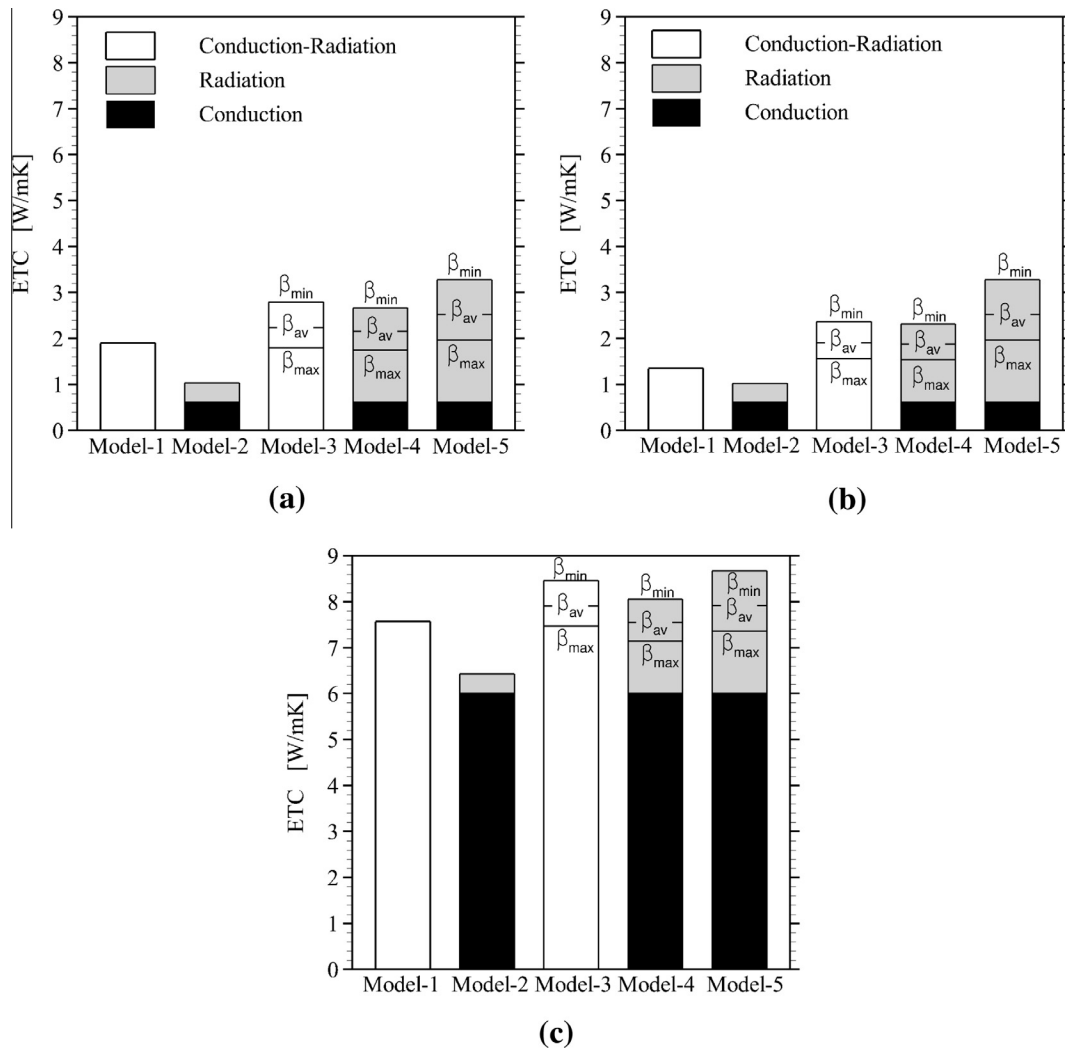


Fig. 4. Effects of different parameters on variations of ETC for Foam-1 evaluated with different models: (a) Case-4 with large $\Delta T = 200$ K, (b) Case-5 with low $\varepsilon = 0.2$ and (c) Case-6 with large $k_s/k_f = 1000$.

Table 4

Deviations (in %) of the predicted ETC for Foam-1, obtained using Models 2 to 5 with respect to Model-1 for all the cases presented in Table 3.

Cases		Model-2	Model-3	Model-4	Model-5
1 (Reference)	$\beta_{\min}$	–45.4	47.7	40.6	73.5
	β_{av}		18.1	13.9	33.7
	$\beta_{\max}$		–5.1	–7.4	4.4
2	$\beta_{\min}$	–50.8	78.9	69.1	115.0
	β_{av}		37.6	31.9	59.5
	$\beta_{\max}$		5.3	2.1	18.6
3	$\beta_{\min}$	–29.0	20.4	16.7	34.0
	β_{av}		4.7	2.5	12.9
	$\beta_{\max}$		–7.6	–8.9	–2.7
4	$\beta_{\min}$	–45.6	47.5	40.5	72.9
	β_{av}		17.9	13.8	33.2
	$\beta_{\max}$		–5.3	–7.6	4.0
5	$\beta_{\min}$	–23.6	75.1	71.5	143.0
	β_{av}		41.9	39.7	87.2
	$\beta_{\max}$		15.8	14.5	46.2
6	$\beta_{\min}$	–15.1	11.9	6.4	14.6
	β_{av}		4.5	–0.2	4.7
	$\beta_{\max}$		–1.3	–5.6	–2.6

not been considered for this comparison since its performance has been already found to be inferior as compared to other models for Foam-1. It is interesting to note that Foams 2 and 3 have higher

porosities (0.74 and 0.79, respectively) as compared to that for Foam-1. It is, therefore, expected that the contribution of radiation should be more prominent for these foams, which can be readily

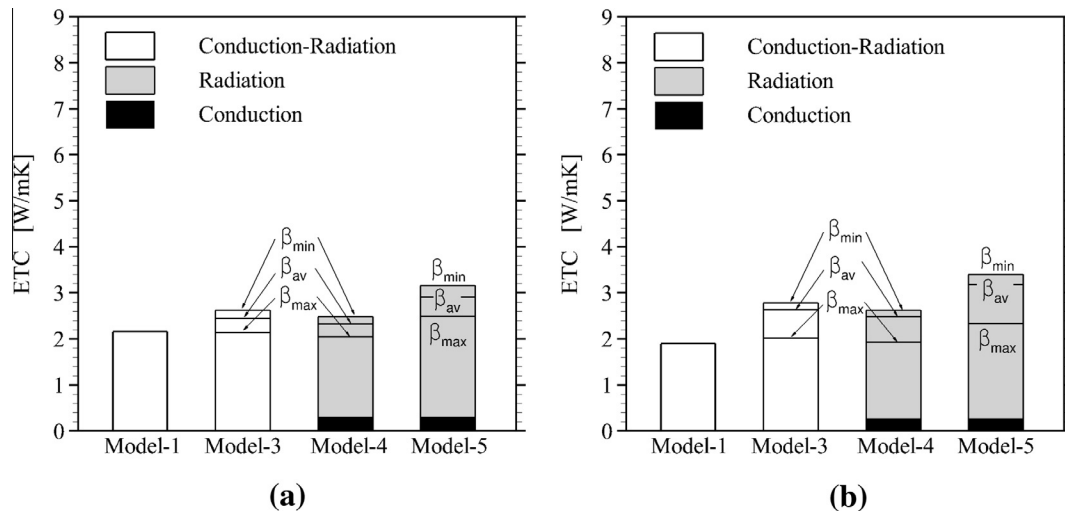


Fig. 5. ETC for case-1 evaluated with different models for (a) Foam-2 and (b) Foam-3.

observed from Fig. 5. Moreover, Fig. 5 also clearly shows that Models 3 and 4 perform much better as usual, whereas Model-5 over-predicts the values of ETC, as observed for Foam-1. However, similar to Foam-1, better performances of these homogeneous models have been observed only when $\beta_{\max}$ has been considered for simulations. The accuracy of these models for Foams 2 and 3, with respect to the reference model (i.e., Model-1), can be quantitatively observed from the data presented in Table 5, which shows that the relative errors yielded by Models 3 and 4, considering the maximum value of the extinction coefficient, are lower than 6% for both foams. Therefore, both these models should be acceptable by considering the simplicity.

3.4. Suggested approach for combined conduction–radiation problems

From the results presented so far, it can be readily concluded that the detailed conduction–radiation model with the blocked-off approach for radiation (Model-1) can be easily applied for complex porous structures, although it still requires experimental validation. In the present investigation, however, results obtained using this approach are considered as the reference since it involves the most detailed numerical simulations. Nevertheless, the requirement of computational time for this reference model is much higher as compared to that required for homogeneous models. For example, with parameters considered in Case-1 for Foam-3, listed in Table 3, the computational time for the detailed reference model (Model-1) using parallel computation with 512 processors (Intel Xeon X5670, 2.93 GHz) has been found to be approximately 34 h, whereas the simulation time for the same condition with homogeneous models using only a single processor (Intel Core i7-2600, 3.4 GHz) has been recorded to be of the order of a few seconds. However, homogeneous models require addi-

tional time for the pre-processing of ETC for conduction, which is approximately 2 h using parallel computation with 8 processors, and additional few seconds for the evaluation of extinction coefficient from 3D CT-scan data using a single processor.

Based on the discussion of present results, an efficient yet accurate method for estimating the ETC of open-cell porous foams at high temperature, relying only upon the information extracted from 3D CT-scan images, can be outlined as follows:

- (i) Evaluate the ETC for pure conduction from 3D detailed simulation. These simulations require considerably less computational time as compared to the combined conduction–radiation problem. The computational effort, however, can be significantly reduced by sacrificing certain amount of accuracy using the methodology proposed by Mendes et al. [8], where the simulation is carried out only once for the solid matrix in order to evaluate the ETC under vacuum condition, while relying upon 3D CT-scan images of the same foam.
- (ii) Evaluate the extinction coefficient using the image processing technique proposed by Loretz et al. [21].
- (iii) Evaluate the ETC using appropriate homogeneous models (either Model-3 or 4) by considering the ETC for pure conduction and the extinction coefficient for radiation heat transfer as inputs, as well as assuming the scattering albedo to be equal to the reflectivity and adopting the scattering phase function from the result of a large opaque diffuse sphere.

It is extremely important to note that although both Models 3 and 4 yield very similar results for the overall ETC, Model-3 should be recommended for being more realistic since it considers the combined interaction between conduction and radiation.

4. Conclusions

In the present study, ETC of open-cell porous foams has been numerically evaluated at high temperatures where both conduction and radiation heat transfer are equally important. Several models, including 3D detailed models and simplified, homogeneous, 1D models have been proposed in the article.

The detailed model is based on the blocked off region approach for radiation, integrated with the finite volume method, for solving

Table 5
Deviation in (%) of the predicted ETC for Foams 2 and 3, obtained using Models 3–5 with respect to Model-1 for Case 1, listed in Table 3.

Sample		Model-3	Model-4	Model-5
Foam-2	$\beta_{\min}$	21.4	14.8	45.9
	β_{av}	13.2	7.5	34.5
	$\beta_{\max}$	–1.1	–5.5	15.2
Foam-3	$\beta_{\min}$	46.3	37.7	78.6
	β_{av}	38.2	30.4	66.9
	$\beta_{\max}$	5.8	1.3	22.9

combined conduction and radiation heat transfer within complex porous structures, as proposed by Talukdar et al. [2]. The most striking feature of this approach lies in the use of voxel-based, uniform, Cartesian grid information that could be easily obtained for a given porous medium from the digitised CT-scan data. Such a simple treatment allows one to predict the interaction of conduction and radiation heat transfer inside the complex structures of porous media, without having the trouble of generating cumbersome numerical grid for the simulations. However, these detailed simulations demand large computational resources as well as time.

The alternative homogeneous models are computationally more economical provided accurate predictions for the ETC due to pure conduction and the extinction coefficient for radiation could be reliably obtained from the CT-scan images of foams. In the present study, the ETC for pure conduction has been calculated from 3D detailed simulation and the extinction coefficient has been evaluated using a simple and fast image processing technique.

In both approaches, the evaluation of ETC (conductive, radiative or combined) is generally based on the detailed structure of the porous foam, which has been extracted from 3D CT-scan images. Three samples of ceramic foams with different porosities and characteristic pore sizes have been considered for the present investigation. In order to evaluate the performance of these models, in-depth parametric study has been carried out for the ETC considering variations in the average temperature, the applied temperature difference between hot and cold boundaries, the emissivity of internal solid surfaces and the conductivity ratio between the solid and the fluid media.

From the present investigation the following main conclusions can be drawn:

- (i) Detailed conduction–radiation model with blocked-off region approach for radiation has been found to be easily implementable as well as applicable for complex porous structures.
- (ii) Calculation of the ETC due to pure radiation using a detailed 3D model and subsequent superposition of the result with the ETC obtained for pure conduction from a separate detailed 3D simulation may not produce the equivalent ETC as compared to that obtained from a combined 3D conduction–radiation model. The deviation has been found to increase with the increase in emissivity of internal solid surfaces.
- (iii) Homogenous models with the solution of radiative transfer equation, either coupled or superposed with conduction, yield quite similar predictions, whereas the homogeneous model with Rosseland approximation slightly over-predicts the value of ETC.
- (iv) Accuracy of homogeneous models strongly depends on the selected value of extinction coefficient, evaluated using the 3D CT-scan image processing technique. In this regard, the maximum value of extinction coefficient in the converged depth of the sample has been found to be the most suitable for a better agreement of the ETC while comparing the result with the detailed reference model.
- (v) Using the maximum value of extinction coefficient, homogeneous models, except that with the Rosseland approximation, yield a relative error that is lower than 10%. However, for lower emissivity of the internal solid surfaces, this error may increase up to 15% since accurate modelling of the scattering phase function could play a very important role.
- (vi) Finally, taking into account the requirement of computational time and effort as well as the accuracy of predicted ETC in presence of conduction and radiation, Models 3 and 4 are recommended for further practical use.

In the end, it may be highlighted here that the present investigation is restricted only to numerical simulations, validation of different theoretical models and study of relative performance of these models. Based on this study, definite recommendation for reliable modelling has been also made. As the next step, experimental validation of the recommended models should be carried out in order to draw more authentic conclusions.

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References

- [1] R. Coquard, D. Rochais, D. Baillis, Conductive and radiative heat transfer in ceramic and metal foams at fire temperatures, *Fire Technol.* 48 (2012) 692–733.
- [2] P. Talukdar, M.A.A. Mendes, R.K. Parida, D. Trimis, S. Ray, Modelling of conduction–radiation in a porous medium with blocked-off region approach, *Int. J. Therm. Sci.* 72 (2013) 102–114.
- [3] J. Kuhn, H.P. Ebert, M.C. Arduini-Schuster, D. Buttner, J. Fricke, Thermal transport in polystyrene and polyurethane foam insulations, *Int. J. Heat Mass Transf.* 35 (1992) 1795–1801.
- [4] T. Dillard, F. Nguyen, E. Maire, L. Salvo, S. Forest, Y. Bienvenu, J.D. Bartout, R. Dendievel, P. Cloetens, 3D quantitative image analysis of open-cell nickel foams under tension and compression loading using X-ray microtomograph, *Philos. Mag.* 85 (2005) 2147–2175.
- [5] M.F. Modest, *Radiative Heat Transfer*, Academic press, New York, 2003.
- [6] J. Petrasch, B. Schrader, P. Wyss, A. Steinfeld, Tomography-based determination of the effective thermal conductivity of fluid-saturated reticulate porous ceramics, *ASME J. Heat Transfer* 130 (2008) 032602.
- [7] R. Coquard, D. Baillis, Numerical investigation of conductive heat transfer in high-porosity foams, *Acta Mater.* 57 (2009) 5466–5479.
- [8] M.A.A. Mendes, S. Ray, A. Trimis, A simple and efficient method for the evaluation of effective thermal conductivity of open-cell foam-like structures, *Int. J. Heat Mass Transfer* 66 (2013) 412–422.
- [9] O. Hahn, F. Raether, M.C. Arduini-Schuster, J. Fricke, Transient coupled conductive/radiative heat transfer in absorbing, emitting and scattering media: application to laser-flash measurements on ceramic materials, *Int. J. Heat Mass Transfer* 40 (1997) 689–698.
- [10] R. Coquard, D. Rochais, D. Baillis, Experimental investigations of the coupled conductive and radiative heat transfer in metallic/ceramic foams, *Int. J. Heat Mass Transf.* 52 (2009) 4907–4918.
- [11] R. Coquard, J. Randrianalisoa, S. Lallich, D. Baillis, Extension of the flash method to semitransparent polymer foams, *ASME, J. Heat Transfer* 133 (2011) 112604.
- [12] C.Y. Zhao, T.J. Lu, H.P. Hodson, Thermal radiation in ultra-light metal foams with open cells, *Int. J. Heat Mass Transf.* 47 (2004) 2927–2939.
- [13] C.Y. Zhao, T.J. Lu, H.P. Hodson, J.D. Jackson, The temperature dependence of effective thermal conductivity of open-celled steel alloy foams, *Mater. Sci. Eng.* 367 (2004) 123–131.
- [14] S.L. Hayes, K.L. Peddicord, Radiative heat transfer in porous uranium dioxide, *J. Nucl. Mater.* 202 (1993) 87–97.
- [15] V. Skibina, Bestimmung der wärmeleitfähigkeit von nichtdurchströmten zellulären metallen, Ph.D Thesis, TU Bergakademie Freiberg, Germany, 2012.
- [16] E. Solórzano, J.A. Reglero, M.A. Rodríguez-Pérez, D. Lehmhus, M. Wichmann, J.A. de Saja, An experimental study on the thermal conductivity of aluminium foams by using the transient plane source method, *Int. J. Heat Mass Transf.* 51 (2008) 6259–6267.
- [17] C. Argento, D. Bouvard, A ray tracing method for evaluating the radiative heat transfer in porous media, *Int. J. Heat Mass Transf.* 39 (1996) 3175–3180.
- [18] J. Petrasch, P. Wyss, A. Steinfeld, Tomography-based Monte Carlo determination of radiative properties of reticulate porous ceramics, *J. Quant. Spectrosc. Radiat. Transfer* 105 (2007) 180–197.
- [19] S. Haussener, W. Lipiński, J. Petrasch, P. Wyss, A. Steinfeld, Tomographic characterization of a semitransparent-particle packed bed and determination of its thermal radiative properties, *ASME J. Heat Transfer* 131 (2009) 1–11.
- [20] P. Parthasarathy, P. Habisreuther, N. Zarzalis, Identification of radiative properties of reticulated ceramic porous inert media using ray tracing technique, *J. Quant. Spectrosc. Radiat. Transfer* 113 (2012) 1961–1969.
- [21] M. Loretz, E. Maire, D. Baillis, Analytical modelling of the radiative properties of metallic foams: Contribution of X-ray tomography, *Adv. Eng. Mater.* 10 (2008) 352–360.
- [22] J.C. Chai, H.S. Lee, S.V. Patankar, Treatment of irregular geometries using a Cartesian coordinates finite-volume radiation heat transfer procedure, *Numer. Heat Transfer, Part B: Fundam.* 26 (1994) 225–235.
- [23] P. Talukdar, Discrete transfer method with the concept of blocked-off region for irregular geometries, *J. Quant. Spectrosc. Radiat. Transfer* 98 (2006) 238–248.

- [24] P. Talukdar, Radiative heat transfer for irregular geometries with the collapsed dimension method, *Int. J. Therm. Sci.* 45 (2006) 103–109.
- [25] J.R. Howell, Determination of combined conduction and radiation of heat through absorbing media by the exchange factor approximation, *Chem. Eng. Progr. Symp. Ser.* 61 (1965) 162–171.
- [26] S.V. Patankar, *Numerical Heat Transfer and Fluid Flow*, Hemisphere Publishing Corp, New York, 1980.
- [27] J.C. Chai, S.V. Patankar, Finite volume method for radiation heat transfer, *Advances in Numerical Heat Transfer*, Chapter Four (2000) 109–138.
- [28] P. Talukdar, M. Steven, F.V. Issendorff, D. Trimis, Finite volume method in 3-D curvilinear coordinates with multiblocking procedure for radiative transport problems, *Int. J. Heat Mass Transf.* 48 (2005) 4657–4666.
- [29] N.G. Shah, New method of computation of radiation heat transfer in combustion chambers, Ph.D. Thesis, Imperial College, University of London, England, 1979.
- [30] P.J. Coelho, M.G. Carvalho, A conservative formulation of the discrete transfer method, *ASME Journal of Heat Transfer* 119 (1997) 118–128.
- [31] S.C. Mishra, P. Talukdar, D. Trimis, F. Durst, Computational efficiency improvements of the radiative transfer problems with or without conduction – a comparison of the collapsed dimension method and the discrete transfer method, *Int. J. Heat Mass Transf.* 46 (2003) 3083–3095.
- [32] P. Talukdar, S.C. Mishra, Transient conduction–radiation interaction in a planar packed bed with variable porosity, *Numerical Heat Transfer-Part A* 44 (2003) 281–297.
- [33] P. Talukdar, S.C. Mishra, Analysis of conduction–radiation problem in absorbing, emitting and anisotropically scattering media using collapsed dimension method, *Int. J. Heat Mass Transf.* 45 (2002) 2159–2168.
- [34] M. Emmel, C.G. Aneziris, Development of novel carbon bonded filter compositions for steel melt filtration, *Ceram. Int.* 38 (2012) 5165–5173.